Software Engineering

2. Risk Response Planning

Incident Response Planning

- Operational backbone for containing, mitigating, and recovering from disruptive events
  - IRP is not a static document
  - It is a living, continuously exercised capability that blends technical execution, decision-making discipline, organizational coordination, and regulatory alignment
- A well-designed IRP ensures that when an incident occurs
  - Even under conditions of stress and uncertainty
    - *Teams respond with speed, clarity, and coherence*
    - *High-maturity incident response minimizes damage, reduces recovery time, and prevents recurrence*

Strategy Components

- Threat scenario catalogs
  - Enumerate and characterize probable or high-impact incident types
  - Unlike generic threat lists, scenario catalogs are contextualized to the organization’s technology stack, business model, and regulatory exposure
  - Examples
    - *Cyber: Ransomware, credential theft, insider exfiltration, command-and-control activity*
    - *Operational: Storage array collapse, message-queue saturation, orchestration failure*
    - *Cloud: Region outages, IAM misconfiguration, API rate limit exhaustion*
    - *Data integrity: Corrupted datasets, failed ETL pipelines, unauthorized data manipulation*
    - *Third-party: SaaS outage, vendor breach, supply-chain compromise*

Strategy Components

- Threat scenario catalogs
  - Each scenario includes
    - *Likelihood*
    - *Affected assets*
    - *Early warning KRIs*
    - *Containment strategies*
    - *Escalation requirements*
    - *Expected recovery pathways*

Strategy Components

- Impact pathways
  - Describe how an incident propagates across systems and business services
    - *Enables responders to anticipate secondary and tertiary effects*
  - Example:
    - *A credential compromise → privileged escalation → database exfiltration → downtime → customer impact → regulatory exposure*
  - Mapping these pathways supports
    - *Prioritized containment*
    - *Isolation strategies*
    - *More accurate business impact forecasting*
    - *Better-tuned KRIs and escalation triggers*

Strategy Components

- Decision trees (contain, isolate, failover)
  - Decision trees guide responders through branching choices based on observed conditions
  - Use cases
    - *Contain: Quarantine compromised endpoints, revoke tokens*
    - *Isolate: Remove components from the network, segment traffic*
    - *Failover: Trigger DR site activation, route traffic to secondary cluster*
  - Decision trees reduce decision fatigue and ensure consistent, policy-aligned responses
  - They document SOP decision making logic

Strategy Components

- Communication matrices
  - A communication matrix defines who communicates what, to whom, when, and through which channels
  - Includes
    - *Incident Commander notifications*
    - *Legal/compliance updates for regulated events*
    - *Executive escalations*
    - *Customer and external stakeholder communications*
    - *Media responses (rare but essential in severe events)*
  - Communication matrices prevent
    - *Conflicting messages*
    - *Delays in decision-making*
    - *Regulatory non-compliance*

Strategy Components

- Runbooks and playbooks
  - Runbooks capture low-level technical steps
  - Playbooks capture higher-level sequences and decision logic
  - Example playbooks
    - *“Suspicious Privilege Escalation”*
    - *“Ransomware Detected”*
    - *“Container Node Failure”*
    - *“Data Integrity Compromise”*

Strategy Components

- Runbooks and playbooks
  - Example runbook steps
    - *Revoke temporary IAM tokens*
    - *Snapshot affected volumes*
    - *Apply firewall rule*
    - *Trigger automated SOAR routine*
  - Runbooks and playbooks together ensure repeatability, automation, and operational consistency

Strategy Components

- Clearly defined coordination roles
  - Incident commander
    - *Owns the operational response*
    - *Approves containment and eradication actions*
    - *Manages cross-team coordination*
  - Communications lead
    - *Handles internal and external messaging*
    - *Keeps executives and regulators informed*
    - *Aligns communications with legal standards*
  - Technical lead
    - *Directs hands-on technical staff*
    - *Manages investigation and triage*
    - *Coordinates with SMEs across teams*
  - Clear role definition eliminates confusion during crises and ensures unity of effort

Strategy Components

- RACI Tables
  - RACI (Responsible, Accountable, Consulted, Informed) ensures that:
  - Responsibility boundaries are explicit
  - Decision-makers are identified
  - No task is “owned by nobody” under pressure
  - Stakeholders receive the right level of communication
  - RACI modeling is essential for minimizing coordination failures.

Scenario-Based IRP Development

- Scenario-driven planning
  - Cstomizes response strategies to different categories of incidents
  - Not all incidents behave the same, and responses must be tailored accordingly
- Cybersecurity incidents
  - Examples: malware infections, credential compromise, DDoS attacks
  - Considerations
    - *Evidence preservation for forensics*
    - *Threat intelligence integration*
    - *Regulatory reporting for breaches*
    - *Eradication without losing root cause evidence*

Scenario-Based IRP Development

- Infrastructure failures
  - Examples: database corruption, storage failures, orchestration system collapse
  - Considerations
    - *Rapid failover and redundancy*
    - *Data recovery from validated snapshots*
    - *Architectural review to prevent recurrence*
- Cloud provider disruptions
  - Examples: cloud region outage, IAM misconfiguration, API throttling
  - Considerations
    - *Multi-region failover*
    - *Automated infrastructure rehydration*
    - *Dependency mapping for distributed systems*
    - *CSP (Cloud Service Provider) SLAs and escalation*

Scenario-Based IRP Development

- Data integrity events
  - Examples: corrupted ETL pipelines, unauthorized changes to datasets
  - Considerations
    - *Isolation of suspect pipelines*
    - *Identification of “last known good state”*
    - *High-trust restore procedures*
    - *Validation before services resume*
- Third-party outages
  - Examples: SaaS provider downtime, vendor breach
  - Considerations
    - *Contractual obligations + SLA enforcement*
    - *Rapid invocation of contingency vendors*
    - *Risk acceptance vs. operational workaround decisions*
    - *Monitoring of vendor-provided status and updates*
  - Scenario-based strategies improve response precision and reduce recovery times

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High-Maturity IR Plan

- Cross-functional rehearsal (tabletop + live-fire)
  - Tabletop exercises (TTX)
    - *Execs, engineers, risk teams walk through scenarios to evaluate decision-making*
  - Live-fire exercises
    - *Simulated attacks or controlled faults (e.g., chaos engineering) test real operational resilience*
  - Benefits
    - *Builds reflexes*
    - *Identifies blind spots*
    - *Strengthens team coordination*
    - *Validates playbooks and automation*

High-Maturity IR Plan

- Automated playbooks for rapid initial actions
  - SOAR platforms execute
    - *Log enrichment*
    - *Endpoint isolation*
    - *Credential revocation*
    - *Firewall rule deployment*
    - *Suspicious user lockouts*
    - *Triage information gathering*
  - Automation reduces MTTD and MTTR, especially in high-volume environments

High-Maturity IR Plan

- Decision checkpoints with predefined criteria
  - Checkpoints ensure escalation and containment decisions are made consistently and timely
  - Examples
    - *“If containment efforts fail after 10 minutes → escalate to Incident Commander.”*
    - *“If data integrity is uncertain → failover to last known good snapshot.”*
  - Checkpoints reduce inconsistency and avoid unnecessary delays

High-Maturity IR Plan

- Regulatory alignment (NIST, ISO 27035, FFIEC)
  - High-maturity IR plans align to recognized frameworks
    - *NIST SP 800-61 (Computer Security Incident Handling Guide)*
    - *NIST CSF – Respond & Recover domains*
    - *ISO 27035 (Information Security Incident Management)*
    - *FFIEC guidance for financial operations resilience*
  - Alignment ensures
    - *Compliance*
    - *Audit readiness*
    - *Consistency with industry norms*

High-Maturity IR Plan

- Integration with GRC systems
  - IR integrates with GRC platforms for
    - *Incident documentation*
    - *Root cause analysis tracking*
    - *Evidence management*
    - *Remediation monitoring*
    - *Lessons learned ingestion*
    - *Automated risk register updates*
  - This creates a closed-loop incident-to-risk feedback mechanism

Incident Lifecycle

- NIST/ISO incident lifecycle
  - Gold standard for incident handling across cybersecurity, IT operations, and organizational resilience
    - *Continuous, adaptive loop, not a linear sequence*
    - *Each phase builds the capability to respond faster, more effectively, and with greater precision*
    - *Ensures that incidents are handled systematically, from preparation and detection, through containment and recovery, to lessons learned and strategic improvements*
  - Phases
    - *Preparation*
    - *Detection and Analysis*
    - *Containment*
    - *Eradication*
    - *Recovery*
    - *Post-Incident Review*

Preparation

- Build tools, access pathways, and communication channels
  - Incident responders require rapid access to
    - *SIEM dashboards*
    - *Cloud consoles*
    - *Endpoint detection systems*
    - *Ticketing and GRC platforms*
    - *Forensic tools (memory analysis, log extractors)*
    - *Network segmentation and firewall controls*
    - *Identity management portals*

Preparation

- Build tools, access pathways, and communication channels
- Preparation includes
  - Pre-loading required tools on secured laptops
  - Ensuring responders have elevated just-in-time access
  - Establishing dedicated communication channels
    - *For example: “War Room” Slack/Teams channels, secure voice bridges*

Preparation

- Define playbooks and escalation paths
  - Playbooks define what actions to take
    - *Escalation paths define who must take them and when*
  - Elements include
    - *Automated first-response actions*
    - *Technical and non-technical decision trees*
    - *Required notifications (legal, compliance, PR, regulators)*
    - *Trigger points for escalating severity levels*

Preparation

- Conduct simulation exercises
  - Simulation builds muscle memory and identifies gaps
  - Types
    - *Tabletop exercises (TTX) to execute scenario walkthroughs*
    - *Blue team drills to perform defensive validation*
    - *Red team engagements to perform adversarial testing*
    - *Purple team exercises for joint attack/defense improvement*
    - *Chaos engineering to assess controlled faults in distributed systems*

Detection and Analysis

- Pre-authorize emergency actions
  - Responders often cannot take urgent actions like shutting down a compromised node without approvals
- Preparation includes
  - *Pre-authorizing critical actions during SEV-1 events*
  - *Creating exception procedures for emergency situations*
  - *Ensuring legal and compliance sign-off in advance*
  - This prevents delays that could significantly expand the blast radius

Detection and Analysis

- Detect anomalies
  - Detection sources include
    - *SIEM correlation rules for privilege escalation anomalies*
    - *IDS/IPS for network intrusion signatures*
    - *APM tools for latency spikes, microservice failures*
    - *Cloud logs (e.g., CloudTrail, Azure Activity Logs) for suspicious role changes*
    - *KRI breaches for alerts for early risk indicators*
  - High-maturity environments blend
    - *Signature-based detection*
    - *Behavioral analytics*
    - *Machine learning anomaly detection*

Detection and Analysis

- Confirm and classify the incident
  - Before triggering full IR, responders validate
    - *“Is this a real incident?”*
    - *“What severity should be assigned?”*
    - *“Which playbook applies?”*
  - Classification decisions affect
    - *Escalation paths*
    - *Communication requirements*
    - *Resource allocation*

Detection and Analysis

- Assess scope, blast radius, and affected systems
  - Assessment includes
    - *Identifying compromised or malfunctioning components*
    - *Determining lateral movement (in cyber contexts)*
    - *Understanding business service impacts*
    - *Estimating potential customer exposure*
  - This step determines how far the incident has spread and what is at risk next

Detection and Analysis

- Perform high-level triage and evidence collection
  - Triaging involves
    - *Capturing logs, memory, and system snapshots*
    - *Tagging affected systems for forensic review*
    - *Prioritizing immediate containment actions*
    - *Preserving evidence for legal/regulatory needs*
  - Evidence accuracy is crucial
    - *Poor collection can jeopardize investigations or compliance*

Containment

- Focuses on preventing further damage and enabling root-cause investigation
- Short-term containment
  - Immediate actions that stabilize the environment
    - *Lock or disable compromised accounts*
    - *Isolate infected endpoints*
    - *Block malicious IPs/domains at the firewall*
    - *Throttle or disable affected microservices*
    - *Cut off external data transfers*
  - Short-term actions aim to stop the bleeding.

Containment

- Long-term containment
  - Implemented once the situation stabilizes
    - *Redirecting traffic to redundant instances*
    - *Applying temporary configuration changes*
    - *Deploying compensating controls*
    - *Creating segmented network zones*
    - *Implementing modified IAM policies*
  - Long-term containment maintains functionality while teams prepare for eradication

Eradication

- Removes the root cause
  - Ensures the environment is free from malicious artifacts or corrupted components
- Key eradication activities
  - Remove malware or persistence mechanisms
    - *Delete malicious binaries*
    - *Remove scheduled tasks, cron jobs, startup scripts*
    - *Neutralize command-and-control channels*
  - Patch exploited vulnerabilities
    - *Apply vendor patches or hotfixes*
    - *Update IAM roles and permissions*
    - *Close misconfigured ports or unnecessary services*
    - *This stops adversaries from re-exploiting the same path*

Eradication

  - Disable malicious identities or processes
    - *Rotating passwords and keys*
    - *Revoking OAuth or API tokens*
    - *Terminating rogue sessions*
    - *Disabling compromised service accounts*

- Eradication focuses on permanent removal of the threat, not just temporary containment

Recovery

- Recovery restores full service and returns systems to a trusted operational state
- Key recovery activities
  - Restore services from backups or replicas
    - *Utilize DR replicas, snapshots, or multi-region failover*
    - *Recover corrupted datasets from known-good backups*
    - *Restart affected services in clean environments*

Recovery

- Key recovery activities
  - Validate data integrity
  - Critical for
    - *Financial transactions*
    - *Healthcare records*
    - *Customer-facing data*
    - *Security logs used for investigations*
  - Validation techniques
    - *Checksums*
    - *Hash comparisons*
    - *Database consistency checks*
    - *Business-level reconciliation (e.g., transaction matching)*

Recovery

- Key recovery activities
  - Enhance monitoring post-recovery
  - Immediately after restoration, organizations increase monitoring to detect
    - *Reinfection attempts*
    - *Post-recovery performance degradation*
    - *Recurrence of suspicious patterns*
  - This “hyper-care” phase ensures that recovery is stable and secure

Post-Incident Review

- Post-incident review (PIR)
  - Transforms incidents into institutional learning opportunities
  - Strengthens long-term resilience
- Identify control failures and systemic weaknesses
  - PIR evaluates
    - *Which controls failed or were bypassed*
    - *Whether escalation was timely*
    - *Whether automation executed correctly*
    - *Which organizational gaps slowed response*

Post-Incident Review

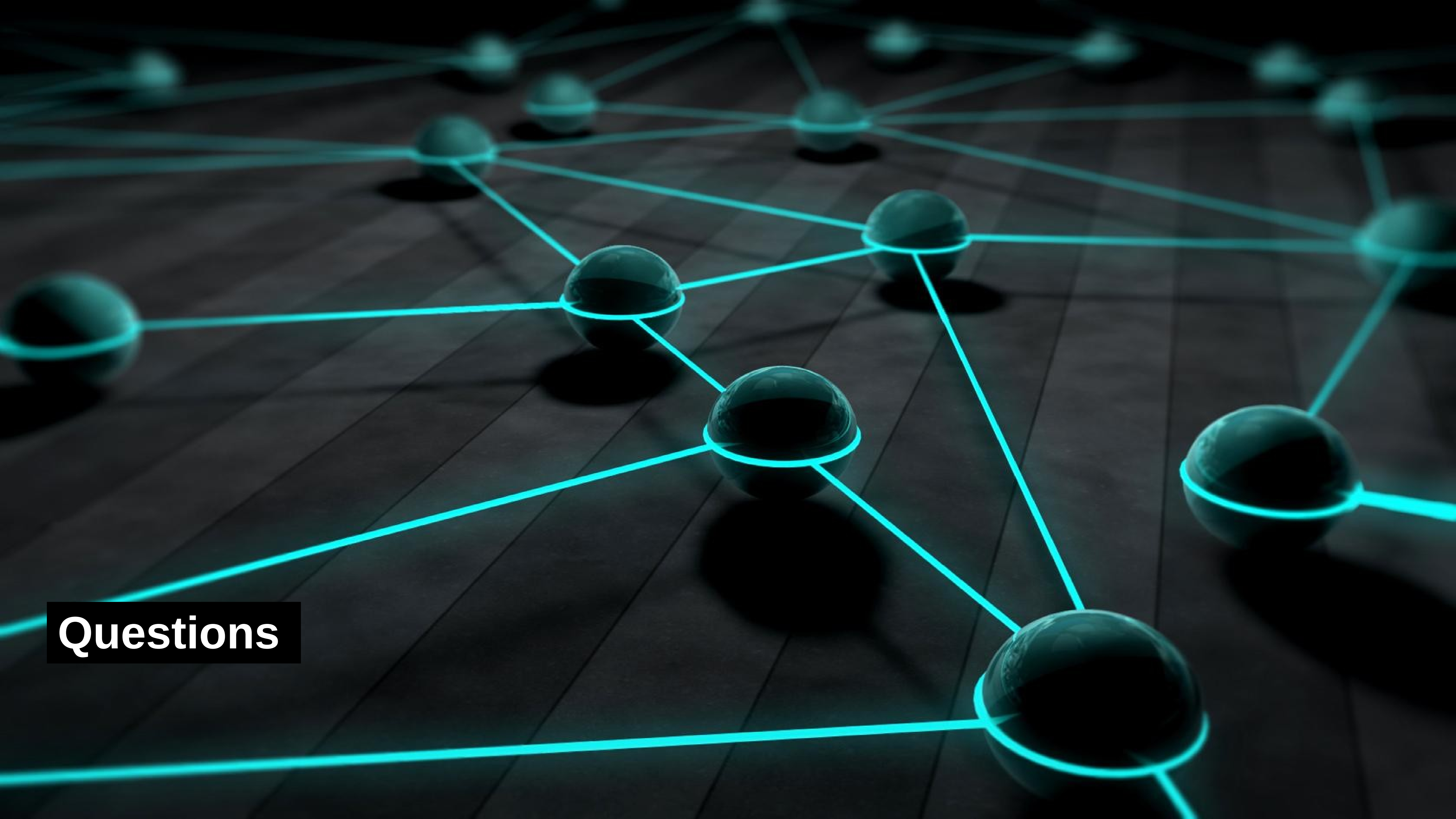
- Update KRIs, controls, and response plans
  - Outcomes may include
    - *Adding new KRIs*
    - *Adjusting KRI thresholds*
    - *Strengthening technical controls*
    - *Revising playbooks*
    - *Improving communication workflows*
  - This ensures the IR program evolves with new insights

Post-Incident Review

- Document the event for compliance and governance
  - Documentation includes
    - *Timeline of events*
    - *Decisions made and rationale*
    - *Root cause analysis*
    - *Evidence collected*
    - *Lessons learned*
    - *Regulatory notifications made*
  - Regulators, auditors, and executive committees require traceability

Post-Incident Review

- Feed insights into future resilience and risk mitigation
  - Outputs integrate into
    - *GRC systems*
    - *Risk registers*
    - *Strategic risk mitigation plans*
    - *Training programs*
    - *Architecture improvements*
  - The goal is continuous improvement and prevention of recurrence



Questions